

The Coordination Cost of Light-Speed Delay in a Self-Replicating Probe Swarm Scales as v/c

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Abstract—A swarm of self-replicating probes fills a stellar field star by star, each probe settling a star and launching copies. The standard exploration-timescale model grants every probe perfect, instantaneous knowledge of which stars are already settled, and names finite light-speed as its own future work. We add that light-speed limit and measure its cost. The first finding is a caution: a model advanced in a timestep coarser than the interval between probe launches makes many probes decide from the same stale snapshot at once, and the resulting collisions appear to slow the fill by tens of percent. That number is a discretization artifact. Resolved so that each probe decides at its true arrival time, the tens-of-percent slowdown collapses – to a few percent at directed-energy speeds and to nothing at slingshot speeds – and the gravitational-slingshot speed-up survives, reproducing the source model quantitatively at about 166 times powered flight. The dominant cost is redundant travel: with essentially the same number of probes built, stale views send probes to stars others have already claimed. We show, and derive from a short collision argument, that this wasted-journey fraction equals the probe’s speed in units of light, v/c – negligible for a powered cruise, about a percent at slingshot speeds, and a fifth at the directed-energy speeds beamed propulsion targets. The tax grows with the replication branching factor before saturating, does not grow with system size over the range we test, and is not a hard floor: a probe that listens in flight, rather than only at its decision stars, recovers essentially all of it at a small cost in fill time. Whether the wasted travel also costs energy is mission-dependent, since a slingshot probe’s kinetic energy is largely gravitationally sourced, so we denominate the tax in journeys. The coordination cost a perfect-information model hides is dominated by fuel and effort and scales, cleanly, as v/c .

Index Terms—self-replication, interstellar exploration, light-speed delay, multi-agent coordination, swarm dynamics

I. INTRODUCTION

A self-replicating probe does not explore a galaxy alone. It settles a star, builds copies of itself from local material, and sends them on to further stars, so a single launch grows into a front that sweeps outward and fills the reachable field. That a machine can build a copy of itself at all is von Neumann’s result, and a probe of this kind saturating the Galaxy in a few million years is the sharpest form of the Fermi question: if interstellar settlement is possible it should already be complete, so the observed silence is a paradox [1], [2]. How long the filling takes is the exploration-timescale question, and it has a long modelling lineage. Early work framed each interstellar probe as an independent agent acting on priors fixed before launch [3], and treated settlement as a front: a density-dependent diffusion with a travelling-wave solution [4], or a Monte Carlo migration filling the Galaxy

in tens of millions of years [5]. Later simulations made the field concrete, sending probes and subprobes through a defined stellar volume to measure fill times directly [6], [7], and established the seeded Monte Carlo methodology that situates an exploration timescale within a distribution of realizations rather than a single number [8].

The sharpest treatment for self-replicators is the slingshot line. Forgan, Papadogiannakis and Kitching showed that a single probe stealing energy from the motion of the stars it passes, a gravitational slingshot, explores up to two orders of magnitude faster than a probe under its own power [9]. Nicholson and Forgan extended that manoeuvre to self-replicating probes and confirmed the speed-up survives replication, with the surprising result that aiming for the nearest star can beat aiming for the largest boost [10]. The probe speeds that make this regime interesting, from the powered cruise of $3 \times 10^{-5} c$ up toward the 0.1 to 0.2 c of directed-energy or high-velocity concepts [11], [12], are exactly where the next assumption starts to hurt.

That assumption, which Nicholson and Forgan name as a limitation and leave for future work, is that every probe is granted perfect, instantaneous knowledge of which stars are already settled. No probe ever flies to a star another has just claimed, because it always knows. In a real galaxy no such oracle exists: news that a star has been settled can travel no faster than light, so a probe choosing its next target acts on a picture of the field that is, for distant stars, years or centuries out of date. This is the elementary consequence of a finite signal speed, that events have only a partial ordering and a remote event is unknowable until its signal arrives [13].

This paper adds that light-speed limit to the self-replicating slingshot model and measures what it costs. The answer, built from a controlled paired-ensemble experiment, has two parts, and the first is a caution. The cost is easy to overstate: a simulation that advances in a fixed timestep coarser than the interval between probe launches makes many probes decide from the same stale snapshot at once, and they collide far more than they physically would. Read that way the delay appears to slow the fill of the field by tens of percent. That number is an artifact. When the dynamics are resolved so that each probe decides at its own arrival time, the tens-of-percent slowdown collapses to a few percent at most, and to nothing at slingshot speeds, while the slingshot speed-up survives contact with finite information intact, reproducing the source model quantitatively. The dominant cost shows up elsewhere. Every

star still gets settled, and with a negligible change in how many probes are built, well under a percent; what the delay chiefly changes is how many wasted journeys the swarm flies to stars other probes have already claimed. That is a cost paid mostly in fuel and effort rather than time, and a perfect-information model hides it entirely. It is governed cleanly by a single dimensionless number, the probe’s speed in units of the speed of light: negligible for a powered cruise, about a percent of journeys at slingshot speeds, and rising to roughly a fifth at the directed-energy speeds that beamed-propulsion concepts target. We derive from a short collision argument that the wasted-journey fraction should equal Λ itself and confirm it in simulation, show the tax grows with the replication branching factor before saturating, and – because a probe deciding only at its stops is pessimistic – that a probe which listens in flight recovers almost all of it, bounding how much coordination can ever buy back. The cost is dominated by fuel and effort, with a secondary fill-time penalty that emerges only at the highest speeds; and it scales, cleanly, with how fast the swarm flies.

II. THE LIGHT-DELAYED BELIEF MODEL

A. The knowledge gate

The change to the model is one gate on knowledge. A settled star is treated as an omnidirectional beacon that begins emitting in the year it is settled. A probe deciding its next target at star f in year Y believes a distant star i is settled only once that beacon’s light has had time to arrive:

$$\text{settled}(i) + \frac{\text{dist}(f, i)}{c} \leq Y. \quad (1)$$

Under the perfect-information assumption of the source model, (1) collapses to “settled at any non-negative year,” and the model reproduces our perfect-information baseline bit-for-bit; the light-speed term is what introduces stale views. When two probes both believe an unsettled star is the best target they both fly to it; the loser arrives to find it taken, its trip wasted, and must re-target from the even more stale picture it holds on arrival. This is availability chosen over consistency: rather than wait for a single agreed map that finite light-speed makes impossible, each probe acts now on its local view [13]. Stars carry velocity magnitudes that feed the slingshot boost but are held fixed in position, following the source model, so $\text{dist}(f, i)$ in (1) is static and the beacon geometry does not shift under the probe. Physical arrival always reads ground truth: what a probe finds on arrival is the star’s real status, whatever it believed when it aimed.

B. The three flight policies, and the speed axis

We carry three named policies, following the slingshot studies we extend [9], [10]. *Powered nearest-neighbour* flight moves under the probe’s own power to the nearest unsettled star. *Nearest-star slingshot* flight takes the same nearest target but gains speed from gravitational slingshots off the stars it passes. *Maximum-boost slingshot* flight instead selects, among the nearest believed-unsettled stars, the target that yields the largest velocity gain. The gravitational boost self-limits (its

per-encounter gain falls off once the probe outruns the stellar escape speed), so both slingshot policies saturate at a mean speed of a few thousand km/s, a few times 10^{-3} of c . To reach the higher speeds that beamed directed-energy concepts target, 0.1 to 0.2 c [11], [12], a probe must be driven, not slung; we therefore sweep the powered cruise speed as a clean, direct control on v/c , from the source model’s $3 \times 10^{-5} c$ up to 0.2 c . Powered flight thus spans the whole speed axis, while the two slingshot policies sit near $v/c \approx 0.01$.

C. Event-driven resolution, and why it is necessary

The fold can advance either by a fixed timestep or by jumping to the next probe arrival. This is not a mere numerical convenience, and getting it wrong is the single largest source of error in this problem. A fixed timestep Δt processes together every probe that arrives within a window, so all of them decide against the same snapshot of the settled map. When Δt exceeds the interval between launches – which it does badly in the boosted regime, where a probe crosses a parsec in centuries against a source-model step of thousands of years – the window batches many independent decisions into one synchronized burst, and they collide far more often than they would if each decided at its true arrival time. The measured coordination cost then reflects the size of the timestep, not the physics. We therefore run the model event-driven throughout: the fold jumps to the earliest outstanding arrival, advances the clock to it, and processes exactly the probes arriving then. This is the exact $\Delta t \rightarrow 0$ limit, dt-independent by construction, and it is what every number in this paper is computed on. Section III shows directly that the apparent fill-time tax is an artifact of a coarse step and collapses to zero as the step resolves.

D. The governing ratio

How much the gate matters is governed by one dimensionless number: the information delay across a hop divided by the travel time across it. For a hop of length d the delay is d/c and the travel time is d/v , so the ratio is

$$\Lambda = \frac{d/c}{d/v} = \frac{v}{c}, \quad (2)$$

the probe’s speed in units of c , independent of the hop length. A slow probe ($\Lambda \ll 1$) outpaces its own ignorance only trivially, so stale views barely arise; a fast probe carries a large Λ and routinely decides before the relevant news can reach it. At the powered cruise speed $\Lambda \approx 3 \times 10^{-5}$ is negligible.

E. A collision model: why the tax should equal Λ

The same ratio predicts the size of the redundant-travel tax, not just its trend, from a short exposure argument. Fix a probe that has committed to a target star at distance d . Under perfect information it chose a truly unsettled star, so it is wasted only if some other probe claims that star during its flight – an exposure window equal to the travel time d/v . Under the light-speed gate the probe carries an *additional* blindness: the target may already have been claimed up to a light-time d/c before

launch, with the beacon not yet arrived, and the probe would have launched anyway. Its total exposure to an unseen claim is therefore $d/v + d/c$. If claims of a given candidate arrive at some roughly steady local rate, the probability of a wasted trip is proportional to the exposure window, so the ratio of wasted trips with and without the gate is

$$\frac{p_{\text{lag}}}{p_{\text{perfect}}} \approx \frac{d/v + d/c}{d/v} = 1 + \frac{v}{c} = 1 + \Lambda, \quad (3)$$

and the coordination tax – the extra wasted trips as a fraction of the perfect-information baseline – is just $p_{\text{lag}}/p_{\text{perfect}} - 1 \approx \Lambda$. Two features of (3) are worth stating because the simulation independently reproduces both: the hop length d cancels, so the tax cannot depend on how far apart the stars are, and the local claim rate cancels, so it cannot depend on the stellar density. This is a leading-order estimate – it assumes claims of a candidate are independent and locally uniform in time, and it ignores the geometry of which stars are in contention – so it predicts the coefficient of proportionality is one, not its small corrections. Section III shows the measured tax follows Λ with a fitted slope of 0.96, confirming the law and its near-unit coefficient, and confirms the d - and density-independence directly.

F. Experimental design and parameters

The field is a box of stars at a uniform density of one star per cubic parsec, the convention of the source model. Each settled star launches offspring (the replication branching factor): branching is what makes two probes ever contend for a star, so it is the precondition for any coordination effect at all and the parameter most likely to change the answer, so we take two offspring as the default and sweep it explicitly (Section III-D). Three properties keep the measurement honest. First, knowledge is evaluated only at the decision star at the moment of decision; news a probe’s path happens to cross in mid-flight is ignored, which deliberately undercounts what a probe could know and so makes the measured waste a conservative upper bound – we measure the opposite, optimistic bound (a probe that listens in flight) separately in Section III-E. Second, a probe that loses a race re-targets, but only up to a fixed number of times before it is retired; this cap is applied identically under both coordination regimes, so the paired comparison is fair, and the measured tax converges as the cap is raised – at $\Lambda = 0.2$ the fuel tax is 6.3, 12.9, 18.4, 20.4, 20.4% for caps of 2, 4, 8, 16, 32, rising while the cap still truncates live bounce chains and saturating by 16 – so our default of 8 is a near-converged mild lower bound, not a free knob. Third, the whole simulation is a pure, seeded, event-driven fold: fixing the seed fixes the star field, every arrival, and every target choice, so a run with the light-speed gate and a run without it can be compared on the *identical* galaxy. That paired design attributes a difference to the delay alone rather than to the luck of the star layout.

Two dependencies are worth stating because they are easy to over-read. Changing the value of the uniform density rescales every inter-star distance by a common factor, multiplying

travel time and light-travel time on every hop equally, so Λ and the relative penalties are unchanged: this is a scale-free property of a uniform field, exactly what (3) predicts when the hop length cancels, so confirming it holds to the printed digit from 0.14 to 5 stars per cubic parsec [14] is a consistency check on the derivation, not an independent robustness result. The case that would genuinely test density sensitivity is a clumpy, non-uniform field, which we do not simulate (a documented limitation). And because the field always fills under both regimes and each settlement launches a fixed number of offspring, the two regimes build essentially the same number of probes – they differ only in a handful of terminal launches as targets run out, well under a percent – so the resource the delay actually moves is the number of journeys flown, which is why the coordination cost is a travel cost.

G. Baseline validation

By construction the “instant” regime is the $c \rightarrow \infty$ limit of (1), and it reproduces the plain perfect-information fold bit-for-bit. Against Nicholson and Forgan’s published model [10] the event-driven fold reproduces both headline findings *quantitatively*: gravitational slingshots explore about two orders of magnitude faster than powered flight (a factor of about 166 on a 400-star field), and their surprising result that aiming for the nearest star beats aiming for the largest boost on total exploration time holds, while the maximum-boost policy reaches the higher peak speed. A coarse fixed step had quantized the boosted hops and undercut this speed-up to about twenty-fold; resolving the step recovers the full factor, which is the same correction that removes the spurious fill-time tax.

III. RESULTS: THE COORDINATION TAX

We measured the cost as a paired ensemble: 48 seeded galaxies of 500 stars, each filled twice on the *identical* field, once with perfect information and once with the light-speed gate of (1), all at the resolved event-driven timestep. Reporting the distribution across the ensemble follows the seeded Monte Carlo practice of the field [8]; we give the median with a seeded percentile-bootstrap 95% confidence interval. We deliberately do not lean on a significance test on the fuel tax: by the structure of the model a stale view can only make a probe believe a settled star is still open, never the reverse, so the decision-site run can only *add* wasted journeys. The sign is therefore built in; what is informative is the magnitude and its scaling, which the intervals and sweeps below report. (This monotonicity is specific to the decision-site gate: the in-flight-relay variant of Section III-E can *lower* waste below the perfect-information baseline, by aborting doomed hops that perfect knowledge at the stops would not.) For the one genuinely null claim – no fill-time tax at low speed – we do not wave the test away but give an equivalence bound instead (below).

A. There is no fill-time tax; the coarse-timestep one is an artifact

Light-speed lag does not measurably slow the fill of the field. Fig. 1 shows why the literature intuition says otherwise. Run with the source model’s fixed timestep the fill-100% penalty for nearest-star slingshot flight is a median of 30.3%, positive in all 32 seeds – a large, robust-looking tax. It is an artifact of the step. As the timestep is refined the penalty falls monotonically – 27.6%, 25.0%, 12.8%, 5.3% at 2000, 1000, 500, 250 years – and at the event-driven limit it is a median of 0.0%, no longer distinguishable from zero. The coarse step had batched many launches into one window so they decided from the same stale snapshot and collided; resolving the step lets each probe decide at its own arrival time, and the collisions that cost time disappear. The same refinement restores the slingshot speed-up to its full value: against powered flight the event-driven fold reproduces Nicholson and Forgan’s roughly two-orders-of-magnitude figure (about $166\times$ on a 400-star field), where the coarse step had reported only about $20\times$. The celebrated speed-up survives contact with finite information; the fill-time tax does not exist.

B. Why the time cost stays far below the fuel cost

Fuel and time are hit unequally because the fill rate is carried by a large parallel population, and a wasted trip removes only one member of it. Fig. 2 tracks the number of probes in flight as the field fills: exponential branching keeps a median of about 480 probes aloft at the peak of a 500-star fill, and the light-delayed swarm carries essentially the same population as the perfect-information one (peaks of about 470 against 480), because both build the same probes. The front advances at the aggregate rate of hundreds of probes settling stars, and losing a fraction of them to wasted trips lowers that rate only in proportion, damped by the redundancy. So a fuel tax of a fifth of journeys does not become a fifth of the fill time: at $\Lambda = 0.2$ the fill-time tax is a few percent, roughly a third of the fuel tax, and it is present throughout rather than only at the end – a median of 4.2% already at half-fill, 5.1% at 90%, and 6.6% at completion. The population stays near its peak all the way into the final percent (about 480 aloft, and still near that at 99% settled, since branching keeps launching until stars run out; Fig. 2), and nearly identical between the regimes, so the two fills proceed almost in lockstep and the lag accumulates gradually rather than spiking at the tail. The large population does not make the time cost vanish, as a coarser reading might suggest; it holds it to a fraction of the fuel cost. At low Λ , where few trips are wasted, that fraction of a small number is what leaves the fill time statistically indistinguishable from the perfect-information one.

C. The real cost is redundant travel, and it scales as Λ

What the delay changes is how far the swarm flies. Because the field always fills and each settlement launches a fixed number of offspring, both coordination regimes build essentially the same number of probes – differing by well under a percent, in a handful of terminal launches – so the resource the delay

TABLE I
COORDINATION TAX OVER THE 48-SEED PAIRED ENSEMBLE AT THE EVENT-DRIVEN TIMESTEP, POWERED FLIGHT, AS A FUNCTION OF $\Lambda = v/c$: THE REDUNDANT-TRAVEL (FUEL) TAX AS A PERCENT OF THE WASTED JOURNEYS UNDER PERFECT INFORMATION, AND THE FILL-TIME TAX, EACH AS A MEDIAN WITH A BOOTSTRAP 95% CONFIDENCE INTERVAL.

$\Lambda = v/c$	Fuel tax (%)	Fill-time tax (%)
0.01	0.6 [0.5, 0.8]	0.0 [0.0, 0.0]
0.03	2.9 [1.8, 4.1]	0.1 [0.0, 1.3]
0.05	4.2 [3.4, 5.8]	0.3 [0.0, 1.8]
0.10	9.3 [7.4, 11.4]	2.6 [0.9, 4.0]
0.20	19.5 [17.8, 23.3]	6.6 [5.2, 10.5]

moves is journeys, and stale views add wasted ones. Table I and Fig. 3 give the redundant-travel tax for powered flight swept across the speed axis: a median of 0.6% of wasted journeys at $\Lambda = 0.01$, rising to 19.5% at $\Lambda = 0.2$, with the confidence interval clear of zero at every speed and the per-seed spread widening with Λ (Fig. 4).

These points sit on the line the collision model predicted. Fitting the tax as $a\Lambda$ through the origin gives $a = 0.96$, within four percent of the derived coefficient of one, and the raw ratio of wasted trips $p_{\text{lag}}/p_{\text{perfect}}$ tracks $1 + \Lambda$ almost exactly – 1.006, 1.042, 1.093, 1.195 measured against 1.010, 1.050, 1.100, 1.200 predicted by (3) at $\Lambda = 0.01, 0.05, 0.1, 0.2$. So the tax is not merely empirical: the wasted-journey fraction *is* Λ , for the reason the derivation gives, the blind stale window d/c adding to the travel exposure d/v . The small shortfall below the line at the lowest speeds is the leading-order model’s expected error, not a different law, and the same argument explains why the tax is independent of hop length and density (both cancel in (3)), which Section III-F and the density check confirm.

A fill-time tax rides alongside the fuel tax, but only once the swarm is fast, and it is a genuine effect rather than a coarse-step artifact. At low speed it is indistinguishable from zero – a median of 0.0% with a bootstrap interval inside $[0, 1.3]\%$ for $\Lambda \leq 0.03$, i.e. equivalent to no delay at a one-percent margin – but it is unambiguously nonzero by $\Lambda = 0.1$ and reaches a median of 6.6% [5.2, 10.5] at $\Lambda = 0.2$ (Table I). This is why we call the cost *fuel-dominated* rather than time-free: at the directed-energy speeds where the fuel tax is largest, a secondary time cost of several percent emerges with it. Section III-B explains why the time cost stays a fraction of the fuel cost rather than matching it – the fill rate is carried by a large parallel population that individual wasted trips barely dent, so wasting a fifth of journeys slows the fill by only a few percent.

D. Contention scales with the branching factor

The tax exists only because probes contend for stars, and contention is created by replication: each settlement launches offspring, and it is those offspring that can race. The branching factor is therefore the parameter most likely to change the answer, so we sweep it. Fig. 5 shows the fuel tax for two, three and four offspring per settlement. At moderate speed

($\Lambda = 0.05$) it is flat, near 6% regardless of branching. At directed-energy speed ($\Lambda = 0.2$) it grows with branching, from 18.4% at two offspring to 24.9% at three, then saturates at 25.5% at four: more offspring make more simultaneous races, but once enough probes are already contending the marginal collision rate levels off. The default of two offspring is thus not a floor that hides a much larger tax at higher branching; the tax rises modestly and then plateaus, and the whole effect stays within the same order of magnitude.

E. The physical floor: how much survives relaying in flight

The tax reported so far is a conservative upper bound, because a probe here learns only at its decision stars and ignores news its path crosses in flight. The opposite extreme is a probe that listens continuously: the moment the beacon from a now-claimed target overtakes it – at the closed-form time set by the beacon closing on the probe at c while the probe closes on the target at v – it abandons the doomed hop and re-aims from wherever it then is. This “in-flight relay” mode is the optimistic bound on what any relay protocol can recover, and we measure it directly. Its effect is dramatic. Because a beacon travelling at c always overtakes a probe travelling at $v < c$ before the probe arrives, a listening probe essentially never completes a wasted arrival: the count of wasted arrivals falls from about 2,070 under decision-site knowledge to zero at $\Lambda = 0.2$. Redundant travel falls too, and below even the perfect-information baseline – from about 3,580 pc under the light-speed gate, and 2,930 pc under perfect information, to about 1,640 pc – because listening in flight also lets a probe bail out of an “assignment collision” (two probes independently choosing the same open star) that perfect knowledge at the decision site does not prevent. The recovery is not free: the mid-flight detours lengthen the fill slightly, by about 2% at $\Lambda = 0.2$. So the decision-site tax is an upper bound, not a floor: a swarm that relays in flight can recover essentially all of the wasted-arrival tax and some of the collisions perfect information leaves behind, at a small cost in time. This is an information ceiling rather than an engineering estimate – the beacon is idealized as isotropic, unattenuated, and detectable the instant its light arrives, so a real relay working against a finite link budget would recover less. What it establishes is that the decision-site tax is nowhere near a physical floor: most of it is protocol, not physics. What genuinely remains is the partial travel a probe flies toward a doomed star before the beacon reaches it, which grows with Λ as the probe covers more of the hop before the relatively slower news catches up.

F. Does the tax hold at scale?

A tax measured on a few-hundred-star field is only interesting if it survives to larger ones, so we sweep the system size at $\Lambda = 0.2$ over a sixteen-fold range, from 300 to 4,800 stars, with 32 to 48 seeds at every size (Fig. 7). The absolute count of wasted journeys grows with the field, as it must, from a median of about 240 extra trips at 300 stars to about 4,000 at 4,800. As a fraction of the perfect-information waste the tax does

TABLE II
WHERE REAL PROPULSION CONCEPTS SIT ON THE Λ AXIS, AND THE FUEL TAX EACH INCURS. THE POWERED CRUISE IS FAR TOO SLOW FOR LAG TO BITE; GRAVITATIONAL SLINGSHOTS SELF-LIMIT NEAR $\Lambda \approx 0.01$; ONLY BEAMED DIRECTED-ENERGY PROPULSION REACHES THE REGIME WHERE THE TAX IS LARGE.

Propulsion regime	$\Lambda = v/c$	Fuel tax (%)
Powered cruise [10]	3×10^{-5}	≈ 0
Gravitational slingshot [10]	≈ 0.01	≈ 1
Directed-energy / light sail [11]	0.1 to 0.2	9 to 20

the opposite: it holds near 18 to 19% up to about a thousand stars and then declines gently, to 16.3% [15.3, 17.8] at 2,400 stars and 13.1% [11.6, 14.0] at 4,800. With this many seeds the decline is resolved rather than suspected – the confidence intervals at 300 and 4,800 stars do not overlap – so it is a real trend, not finite-sample scatter, though the event-mode $O(N^2)$ cost still caps how far we can push it. The direction is what matters for the one extrapolation a reader might be tempted to make: the fraction does not grow with scale, and in fact shrinks, so nothing here supports carrying a fixed-percentage tax out to a 10^{11} -star galaxy. We report the scale behaviour over the range we can measure and make no claim beyond it.

G. Where the cost falls

Table II places the physical propulsion concepts on the speed axis. The source model’s powered cruise ($3 \times 10^{-5} c$) is far too slow for any race to matter, and pays nothing. The gravitational slingshot that motivated this study reaches only $\Lambda \approx 0.01$ before its boost self-limits, and pays about a percent of journeys – real, but small. The tax becomes large only for beamed directed-energy or light-sail propulsion at 0.1 to 0.2 c [11], [12], where it takes a tenth to a fifth of journeys. Every case still fills the field to completion: this is a property of the model, not a result, since with re-targeting and no loss term the reachable field must close, and delay supplies no term that would leave a permanent unsettled remainder of the kind an “Aurora” steady state requires [12]. The coordination failure shows up as wasted motion, not as an unfilled galaxy.

IV. DISCUSSION

The result refines rather than overturns the slingshot picture, and it does so twice over. Nicholson and Forgan’s speed-up is real, large, and here reproduced quantitatively [10]; finite light-speed does not erode it, and does not measurably delay the fill of the field. What a perfect-information model hides is not lost time but lost motion: probes flying to stars that others have already claimed. That redundant travel is the coordination tax, it is governed by the probe’s speed in units of c , and it matters only once a swarm flies fast enough for a decision to outrun its own news.

A. What the tax is denominated in

Because the field always fills and each settlement launches a fixed number of offspring, the two coordination regimes build essentially the same number of probes, differing by well

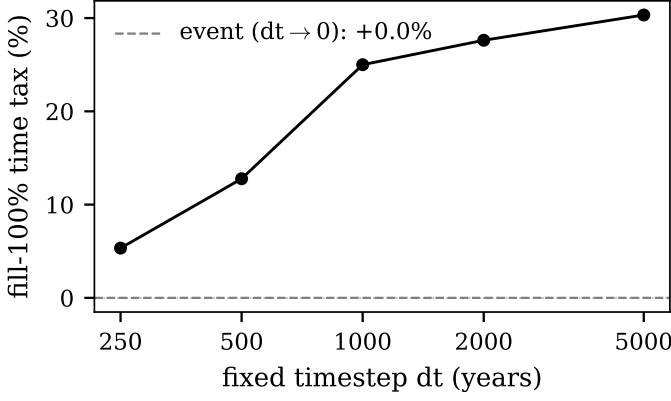


Fig. 1. The fill-100% time penalty for nearest-star slingshot flight versus the fixed timestep, over the 32-seed ensemble. The apparent penalty is monotone in the step and collapses to zero (dashed line: the event-driven, dt-independent limit). The tens-of-percent “coordination tax” of a coarse-step model is a discretization artifact, not a physical slowdown.

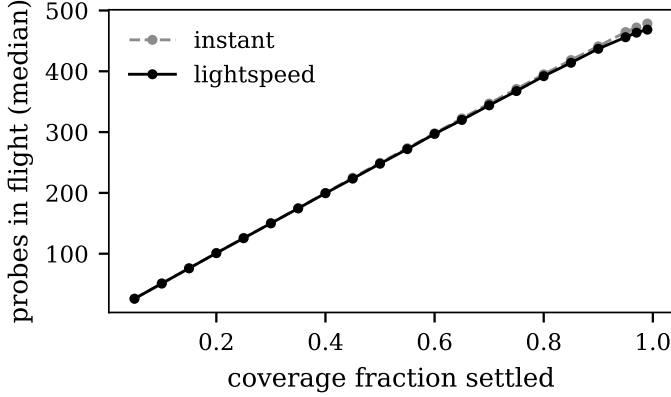


Fig. 2. Probes in flight versus the fraction of the field settled, for perfect information and the light-speed gate (medians over 16 seeds, 500-star field, $\Lambda = 0.2$). The population stays near its peak of a few hundred all the way into the final percent and the two regimes track each other closely, so the fills run almost in lockstep and the light-lag swarm loses only the small productive-rate deficit that keeps the time tax a fraction of the fuel tax.

under a percent; the delay changes almost nothing about how much is manufactured, only how far it is flown. The tax is therefore a travel cost, and we deliberately denominate it in journeys rather than energy, because translating journeys into energy is mission-dependent in a way our model does not fix. A wasted journey’s kinetic energy is not straightforwardly a fuel cost: under the slingshot policies most of that energy was extracted gravitationally from stellar motion rather than spent as propellant, and a probe that re-targets on arrival keeps its speed and pays nothing further. The genuine energy penalty of a wasted trip is whatever a rendezvous at the doomed star would have cost – braking to rest and re-accelerating – which scales with v^2 for a fuel-braked craft but is near zero for a flyby or a gravitationally-braked one. We therefore do not headline an energy figure; the honest statement is that the journey tax becomes an energy tax only to the extent a mission must brake, and then it grows with v^2 . A caveat cuts the other way, too: even as a travel cost the tax is denominated

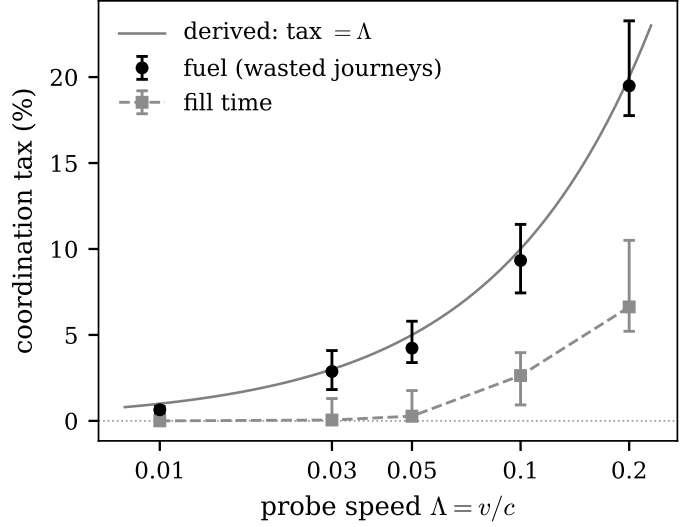


Fig. 3. The coordination tax versus probe speed $\Lambda = v/c$, powered flight, over the 48-seed ensemble at the event-driven timestep (both as a percent over perfect information; points: medians; bars: bootstrap 95% confidence intervals). The redundant-travel (fuel) tax sits on the derived law tax = Λ (grey line, Eq. 3) at every speed; the fill-time tax stays near zero at slingshot speeds and emerges to a few percent only in the directed-energy regime.

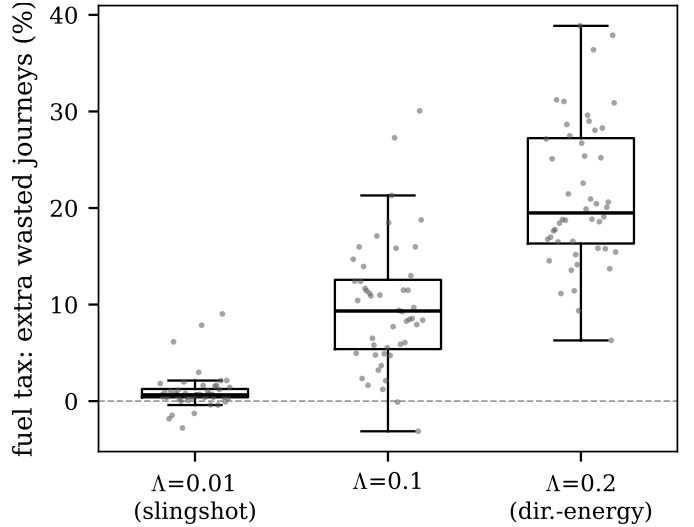


Fig. 4. Per-seed distribution of the fuel tax (extra wasted journeys, as a percent of the perfect-information waste) at slingshot speed ($\Lambda = 0.01$), $\Lambda = 0.1$, and directed-energy speed ($\Lambda = 0.2$), over the 48-seed ensemble (box: interquartile range and median; points: individual seeds). The tax is small but consistently positive at slingshot speed and unambiguous at directed-energy speed.

in journeys, not in the swarm’s total resource budget, which is dominated by building the probes – a cost we hold fixed and identical across regimes and do not model. As a fraction of total mission resources the coordination tax is smaller than its share of journeys.

B. Limitations

The decision-site headline is a conservative upper bound: probes here learn only at their decision stars, ignoring news

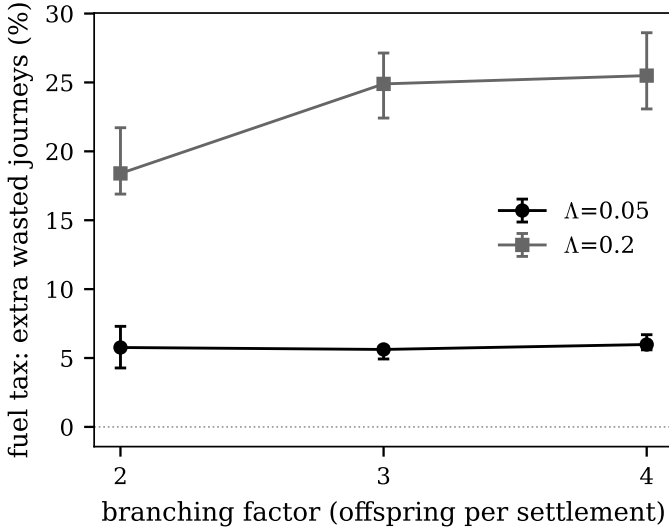


Fig. 5. Fuel tax versus the replication branching factor (offspring per settlement) at moderate and directed-energy speed, over the 32-seed ensemble (points: medians; bars: bootstrap 95% confidence intervals). Contention grows with branching at high Λ and then saturates; the default of two offspring does not hide a much larger tax.

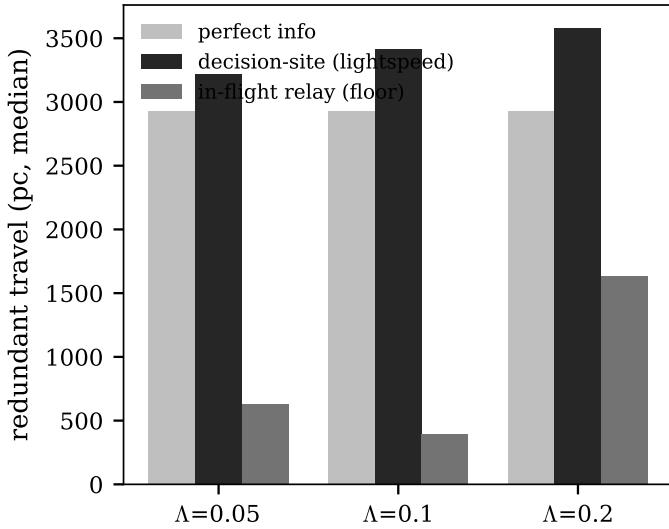


Fig. 6. Redundant travel under perfect information, the decision-site light-speed gate, and in-flight relay, versus Λ (medians over 48 seeds, 400-star field). Listening in flight drives completed wasted arrivals to zero and cuts redundant travel below even the perfect-information baseline, recovering the light-lag tax at a small cost in fill time.

their paths cross in flight. We relax exactly that assumption in Section III-E, where a probe that listens in flight recovers essentially all of the wasted-arrival tax, so the headline is an upper bound rather than a floor. A true probe-to-probe relay of the settled map, which we do not implement, would sit between the beacon-only in-flight bound and the decision-site one and is the natural next step. The floor beneath all of them – that no probe can know a distant star is taken until its light arrives – is set by physics, not protocol.

A further, inherited idealization concerns the stars them-

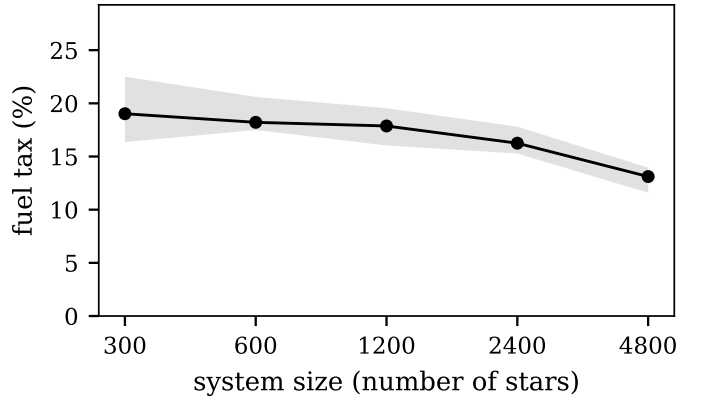


Fig. 7. Fuel tax as a percent of the perfect-information waste versus system size, at $\Lambda = 0.2$ (point: median; shaded band: bootstrap 95% confidence interval; 32 to 48 seeds per point). Over a sixteen-fold range the tax holds near 18 to 19% up to about a thousand stars and then declines to about 13% at 4,800; the end intervals do not overlap, so it does not grow with scale but gently shrinks, which is why we make no fixed-percentage extrapolation to galactic size.

selves. They carry the velocity vectors that power each slingshot but are held fixed in position, exactly as in the two source models, both of which name this as their single most important simplification [9], [10]. It is not a small one: at the local circular speed a star sweeps about 225 pc in a million years, and even the peculiar velocity dispersion alone carries it some 30 to 40 pc [15], [16], far past the roughly one-parsec mean hop, so over a Myr-scale fill real proper motion would reshuffle which stars are near which. One might object that this touches the delayed-belief gate itself, not just the backdrop: the beacon leaves where the star *was*, so a moving field would need retarded positions in (1). But that specific correction is tiny, and for a clean reason: during a beacon's light-crossing a star moves only the fraction v_*/c of the distance it is signalling across, and $v_*/c \approx (40 \text{ km/s})/c \approx 10^{-4}$ – a hundredth of a percent, whatever the decision distance – so the gate reads essentially the right geometry at the moment of decision. What the frozen field removes is the slow, Myr-scale drift of the layout, and that is exactly what the paired design controls: both coordination regimes run on the *identical* frozen field, so the difference still isolates the light-lag cost even where the absolute layout is idealized. The absolute fill times, however, do inherit the idealization.

Three further bounds on scope. First, the field is uniform; changing its density only rescales the clock (confirmed), but a clumpier distribution would lengthen the long-range hops and is a genuine unknown we do not probe. Second, at high v/c the hops are short, so an event-driven run generates of order one arrival per star settled many times over and costs roughly $O(N^2)$; this bounded the size sweep to a sixteen-fold range. Over it the tax does not grow with scale – it holds near a fifth up to about a thousand stars and then trends downward at the largest fields we can reach (Section III-F) – so we make no quantitative extrapolation to galactic ($\sim 10^{11}$ -star) scale. A sixteen-fold lever arm cannot support one, and one over

which the fraction if anything shrinks certainly cannot. Third, the dynamics have no loss term: settled sites never fail, so the field always closes. A settlement-death term of the kind that produces the “Aurora” steady state [12] is what would be needed to turn coordination failure into a permanent unsettled remainder; delay alone does not.

C. Relation to delayed agreement

The behaviour is, by analogy and not by derivation, the signature of delayed multi-agent agreement, and we cite that literature only to place the regime. Consensus protocols degrade as communication delay grows against the decision cadence, staying stable only below a delay bound set by the coupling [17], [18]. We borrow the picture, not the theorems: the classical impossibility results assume fault models that are not ours – asynchronous crash faults for FLP [19], network partition for CAP [20] – and none of them predicts the $\Lambda = v/c$ scaling that is this paper’s result. What they do fix is the regime. At parsec spacing the round-trip light-time is years, far longer than any plausible decision cadence, so a swarm that must keep acting can only trade a single consistent map for availability, which is exactly what (1) encodes. The same delay budget drove the teleoperation and networking communities from closed-loop control through move-and-wait and supervisory control [21], [22] to delay-tolerant store-and-forward once no continuous path survives [23]–[25]; at interstellar hops every faster regime is left behind, and each probe is an independent colony acting on priors. The coordination tax is what that independence costs when probes want the same stars.

D. A design lesson

The corollary is a threshold, not a preference. At slingshot speeds the tax is about a percent of journeys: coordination machinery – relay, gossip, a shared map – is not worth its mass, and a designer should spend everything on speed. At the directed-energy speeds that make a probe fast enough to matter, the same physics that buys the speed also makes stale views expensive, and a fifth of journeys wasted is worth engineering against – and, as Section III-E shows, a probe that merely listens in flight buys back nearly all of it. The faster the swarm you can build, the more that little coordination is worth buying, and the finite speed of light sets only the small residual – the travel already flown before the news arrives – that no protocol can ever recover.

V. CONCLUSION

Adding a finite speed of information to the self-replicating slingshot model turns a convenient assumption into a measurable cost, but not the cost a coarse model reports. Once the dynamics are resolved so that each probe decides at its true arrival time, the tens-of-percent fill-time penalty seen in a fixed-timestep model collapses – it was a discretization artifact – and the slingshot speed-up survives intact, reproducing Nicholson and Forgan quantitatively. The genuine cost is redundant travel: every star is still settled, with a negligible

change in the number of probes built, but stale views send probes to stars others have already claimed. That waste follows a simple law we both derive and measure – the wasted-journey fraction equals the probe’s speed in units of light, $\Lambda = v/c$ – from negligible at the powered cruise, through about a percent at slingshot speeds, to a fifth at the directed-energy speeds beamed propulsion targets. It grows with the replication branching factor before saturating, and whether it also costs energy is mission-dependent: only a rendezvous that must brake pays for a wasted trip’s kinetic energy, which for the fast policies was largely gravitationally sourced rather than fuelled. The cost is dominated by fuel and effort; a secondary fill-time penalty of a few percent rides alongside it, but only in the fast directed-energy regime and never at slingshot speeds. It is not a hard floor: a probe that listens in flight, one of the two extensions the model invites, recovers essentially all of the wasted-arrival tax at a small cost in fill time, while a full probe-to-probe relay of the settled map remains open. Neither can remove the physical floor beneath the tax, and how much either can buy back is itself bounded by the finite speed of light.

DATA AVAILABILITY

The model is a deterministic, seeded, event-driven fold; every figure, table, and reported statistic regenerates from source by running the swarm module’s experiment and figure scripts, and every quantitative claim traces to a cited source in the project’s shared bibliography. The code and its citation metadata are archived with the concept DOI 10.5281/zenodo.21249296.

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